

Homework 3

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1. Let G be a group and let H be a subgroup of G . Prove that $C_G(H) \subset N_G(H)$. Prove that equality holds if H has order two.

Proof. If $g \in C_G(H)$, then $ghg^{-1} = h$ for each $h \in H$, so $gHg^{-1} = H$. Thus $g \in N_G(H)$.

If H has order two, then one of the two elements of H is e , and the other is an element of order two, say x . If $g \in N_G(H)$, then $geg^{-1} = e$, and g must act on x to take it to the only remaining element of H , which is x itself, so $gxg^{-1} = x$. $\square$

2. Let G be a group, and suppose H is a finite subgroup of G generated by a set X . Prove that $g \in N_G(H)$ if and only if $gxg^{-1} \in H$ for all $x \in X$. Conclude that for $x \in G$, one has that $g \in N_G(\langle x \rangle)$ if and only if $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$.

Proof. If $g \in N_G(H)$, then $gHg^{-1} = H$, so indeed $gxg^{-1} \in H$ for any $x \in X \subset H$.

Now let $g \in N_G(H)$ and assume that $gxg^{-1} \in H$ for all $x \in X$. Then given $h \in H$, we can write $h = gh'g^{-1}$ for some $h' \in H$, so $h' = x_1^{a_1} \cdots x_n^{a_n}$ for some $x_i \in X$ and each $a_i \in \mathbb{Z}$. We then write

$$h = gh'g^{-1} = gx_1^{a_1}x_2^{a_2}\cdots x_n^{a_n}g^{-1} = (gx_1g^{-1})^{a_1}(gx_2g^{-1})^{a_2}\cdots(gx_ng^{-1})^{a_n},$$

so h is a product of elements of H . Thus $gHg^{-1} \subset H$.

Now assume that $h, h' \in gHg^{-1}$. Then

$$ghg^{-1} = gh'g^{-1} \implies h = h'.$$

Thus since H is finite and $f : H \rightarrow H$ defined by $f(h) = ghg^{-1}$ is injective, f is in fact a bijection, and $gHg^{-1} = H$. $\square$

3. Let $H = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \subset \text{GL}_2(\mathbb{R})$, and let $g = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Show that $gHg^{-1} \subset H$, but that this containment is proper and hence g is *not* in the normalizer of H . Why does this not violate the statement in Problem 2?

Proof. Since any element of H is of the form $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ and $g^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$, we can calculate explicitly

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}.$$

Since this only includes matrices with an even upper-right entry, it excludes $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ itself, and thus $gHg^{-1} \subsetneq H$. This is in complete agreement with Problem 2, since Problem 2 required the subgroup in question, here H , to be finite, and H is infinite. $\square$

4. Let G be a group, and n a positive integer. Prove that the set of group homomorphisms from $\mathbb{Z}_n$ to G are in bijection with the set of elements of G of order dividing n .

Proof. We will write $\mathbb{Z}_n = \langle x : x^n \rangle$ for ease of notation. The idea here is that a homomorphism from $\mathbb{Z}_n$ to G is uniquely determined by where it sends x . Indeed, given a homomorphism φ which sends x to $g \in G$, then $\varphi(x^i) = \varphi(x)^i = g^i$, so φ is described completely by where it sends x since x generates $\mathbb{Z}_n$. A direct consequence is the injectivity we need. Then, using the fact that $|\varphi(x)|$ divides $|x| = n$, we determine that we can only send x to elements of G with order dividing that of x . In fact, given any $g \in G$ with $|g|$ dividing n , there is a homomorphism determined entirely by sending x to g , so we have surjectivity. $\square$

5. A permutation $\sigma \in S_n$ of the form $(i \ j)$ for $1 \leq i < j \leq n$ (in cycle notation) is called a *transposition*. If $j = i + 1$, we say the transposition is *adjacent*. Prove that S_n is generated by the set of transpositions. Then prove that all transpositions are products of adjacent transpositions to conclude that S_n is in fact generated by adjacent transpositions.

Proof. First, we show that every element of S_n is a product of transpositions. Indeed, for a cycle $(a_1 \ a_2 \ \cdots \ a_m)$, we can write

$$(a_1 \ a_2 \ \cdots \ a_m) = (a_{m-1} \ a_m) \cdots (a_2 \ a_m)(a_1 \ a_m).$$

Given any transposition, then, we want to show that it is a product of adjacent transpositions, which we will prove by induction. We write any transposition as $(a \ [a + i])$. Let $\sigma \in S_n$. If $i = 0$, then σ is the identity permutation, which is the product of zero adjacent transpositions. If $i = 1$, Then $\sigma = (a \ [a + 1])$ is an adjacent permutation by definition. Now assume that $i = n$ and the hypothesis holds for $i < n$. Then we can write

$$(a \ [a + n]) = (a \ [a + n - 1])([a + n - 1] \ [a + n])(a \ [a + n - 1]),$$

and since $(a \ [a + n - 1])$ can be written as a product of adjacent permutations, this represents $(a \ [a + n])$ as a product of adjacent permutations. $\square$

6. Let $p \geq 3$ be a prime integer. Prove that D_{2p} is generated by any pair of distinct reflections. Show by means of example that this can fail if p is not prime.

Proof. NOT STARTED $\square$

7. Let G be a group and let H be a subgroup. Let G act on the set G/H by left translation. Prove that the kernel of this action is $K = \bigcap_{g \in G} gHg^{-1}$, and prove that K is the largest normal subgroup of G contained in H .

Proof. NOT STARTED $\square$

8. Suppose G acts on a set X , let $x \in X$, and let $H = \text{Stab}_G(x)$. Recall that G/H denotes the set of left cosets of H in G . Prove that the assignment

$$\begin{aligned}\varphi : G/H &\rightarrow \text{Orb}_G(x) \\ gH &\mapsto g \cdot x\end{aligned}$$

is a well-defined bijective function. Furthermore, show that φ is G -equivalent; that is, prove that for all $a \in G$ and all $gH \in G/H$, one has that $\varphi(a \cdot gH) = a \cdot \varphi(gH)$, where the G -action on G/H is left translation of cosets, and the G -action on $\text{Orb}_G(x)$ is inherited from the action of G on X .

Proof. **NOT STARTED** □

9. Let $\varphi : G \rightarrow H$ be a group homomorphism. Prove that if N is a subgroup of H then $\varphi^{-1}(N)$ is a subgroup of G . Prove that if N is normal, so is $\varphi^{-1}(N)$.

Proof. Let $g, h \in \varphi^{-1}(N)$. Then $\varphi(gh^{-1}) = \varphi(g)\varphi(h)^{-1} \in N$, so $gh^{-1} \in \varphi^{-1}(N)$.

Now given $g \in G$ and $n \in \varphi^{-1}(N)$, we have $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} \in N$ since $\varphi(n) \in N$ and N is normal. Thus $gng^{-1} \in \varphi^{-1}(N)$, so since g and n were arbitrary elements of G and $\varphi^{-1}(N)$ respectively, $\varphi^{-1}(N)$ is normal. □

10. Suppose that G is a group and that X is a subset of G that is closed under conjugation. Prove that $\langle X \rangle$ is a normal subgroup. Next, let Y be the set of commutators

$$\{aba^{-1}b^{-1} : a, b \in G\}$$

and prove that Y is closed under conjugation. Use these facts to show that the commutator subgroup $[G, G] = \langle Y \rangle$ is a normal subgroup.

Proof. If $n \in \langle X \rangle$ and $g \in G$, write $n = n_1 n_2 \cdots n_m$ where each $n_i \in X$. Then

$$gng^{-1} = gn_1 \cdots n_m g^{-1} = (gn_1 g^{-1}) \cdots (gn_m g^{-1}) \in \langle X \rangle$$

since $gn_i g^{-1} \in X$ for each i .

Now, given any $a, b, g \in G$, we have

$$g(aba^{-1}b^{-1})g^{-1} = (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1} \in Y.$$

Thus $[G, G] = \langle Y \rangle$ is a normal subgroup. □